

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** designing and owning distributed systems at scale across **Google, YouTube, Microsoft, and Whatnot**. Built core infrastructure serving billions of daily requests, led teams of 10+ engineers, and managed reliability across multi-region **Kubernetes** clusters. Implemented critical security remediation affecting 15,000+ employee devices and reduced P0 incidents through systematic on-call process improvements. Seeking to bring deep infrastructure and systems expertise to a team Works on network configuration and edge cloud platform reliability. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS

Distributed Systems: Go, Python, C++, Kubernetes, Istio, gRPC, Protocol Buffers, Bigtable

Infrastructure: Borg, Terraform, Google Cloud Platform, Azure, Linux, Datadog, Prometheus, PAVE

Storage: SQL, MySQL, PostgreSQL, Redis, Cassandra, S3, Object Storage

Networking: HTTP/2, QUIC, CDN, TCP/IP, Load Balancing, Network Security

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Own end-to-end video delivery infrastructure serving 2+ billion monthly users, designing systems that process millions of concurrent streams with low latency requirements.
- Lead technical architecture decisions for new transcoding pipeline features, coordinating across video, audio, and CDN teams to meet quality and performance targets.
- Define reliability SLOs for critical playback services, implementing automated alerting that reduced customer-facing incidents by identifying degradation before user impact.
- Collaborate with product teams to ship features improving viewer engagement metrics, validating designs through A/B testing frameworks at massive scale.
- Mentor senior engineers on distributed systems patterns, establishing design review processes that improved cross-team alignment on infrastructure initiatives.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Architected migration of monolith services to microservices using **Go** and **Kubernetes**, reducing deployment cycle time from days to under two hours.
- Established on-call rotation and incident response procedures that decreased mean time to resolution for P0 issues by 40% during first quarter.
- Designed real-time bidding and auction systems for a live commerce platform, processing thousands of concurrent transactions during peak streaming events.
- Led technical hiring efforts, conducting system design interviews and calibrating evaluation rubrics that improved candidate quality and diversity outcomes.
- Built notification infrastructure handling multi-channel delivery (email, push, SMS) with fallback mechanisms ensuring 99.9% message delivery rates.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Managed team of 10+ engineers building infrastructure for Google's internal developer platform, prioritizing roadmap items that shipped features used by 25,000+ engineers globally.
- Designed and shipped PAVE, an internal deployment platform handling millions of daily releases across **Borg** clusters with zero-downtime deployment strategies.

- Led security remediation project implementing endpoint verification for 15,000+ corporate devices, reducing unauthorized access incidents by 95% within six months.
- Owned **Bigtable** infrastructure supporting search indexing workloads, implementing schema optimizations that decreased read latency by 30% for time-sensitive queries.
- Conducted performance reviews and career development for 12 engineers, promoting 4 to senior level and establishing mentorship pairing programs.

Software Engineer | Google | 02/2011 - 02/2015

- Built monitoring and observability tools for **Borg** cluster management, visualizing system health across thousands of machines in real-time dashboards.
- Participated in 24/7 on-call rotation for critical infrastructure services, resolving P0 incidents affecting Google Search and Ads serving pipelines.
- Developed automated capacity planning algorithms that predicted resource needs 72 hours ahead, improving cluster utilization efficiency by 25%.
- Implemented automated remediation scripts for common infrastructure failures, reducing on-call engineer intervention for recurring issues by 70%.
- Contributed to **Kubernetes** upstream project, submitting patches for networking plugins that improved pod-to-pod communication reliability.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Designed and implemented backend services for **Azure** storage platform, building replication logic that maintained data durability across distributed datacenter failures.
- Built diagnostic tools for **SQL Azure** that identified query performance bottlenecks, helping customers reduce database costs by 20-40%.
- Developed **C++** components for Windows kernel networking stack, implementing optimizations that reduced packet processing overhead by 15%.
- Authored technical documentation and API specifications used by 5,000+ external developers integrating with Azure services.
- Participated in design reviews ensuring new features met security and compliance requirements for enterprise customers.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed automated testing framework for **SQL Server** query optimizer, identifying edge cases that improved release stability.
- Collaborated with senior engineers on database indexing algorithms, contributing to performance improvements validated through benchmarking.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007